1. You roll a fair four sided dice, and then a fair six sided dice. You add the numbers on the two dice. What is the probability the result is even?

```
assuming that each dice contains one side for each natural number in
ascending order, the probability is 1/2. Some code to show this:
```

```
even_count = total_count = 0
for i in range(4):
    for j in range(6):
        if (i + j) % 2 == 0:
            even_count+=1
        total_count+=1
print(f"even ratio: {even_count}:{total_count}")
```

```
even ratio: 12:24
```

2. You shuffle a standard deck of playing cards and draw a card.

- (a) What is the probability that this is a king?

```
4 kings in a 52 card deck: 4/52 or 1/13
```

- (b) What is the probability that this is a heart?

```
13 hearts in a 52 card deck: 13/52 or 1/4
```

- (c) What is the probability that this is a red card (i.e. a heart or a diamond)?

```
26 hearts in a 52 card deck: 26/52 or 1/2
```

3. A roulette wheel has 36 slots numbered 1–36. Of these slots, the odd numbers are red and the even numbers are black. There are two slots numbered zero, which are green. The croupier spins the wheel, and throws a ball onto the surface; the ball bounces around and ends up in a slot (which is chosen fairly and at random).

These answers were made with the assumption that the two green spots do NOT contribute to the total count of 36 slots and the total number of slots is 38

- (a) What is the probability the ball ends up in a green slot?

```
2/38 or 1/19
```

- (b) What is the probability the ball ends up in a red slot with an even number?

```
0
```

- (c) What is the probability the ball ends up in a red slot with a number divisible by 7?

```
3/38 - the three numbers this could be: 7, 21, and 35.
```

4. You remove the king of hearts from a standard deck of cards, then shuffle it and draw a card.

- (a) What is the probability this card is a king?

```
3/51 or 1/17
```

- (b) What is the probability this card is a heart?

```
12/51 or 4/17
```

5. Refer to the student alcohol data in Homework #2, Problem 1.14. Let us focus on the students who took math (file: student.math.csv), and their final grade (G3) and absences. The final grade (G3) is on a scale 0-20. Convert that to a percentage score, and assume that grade A is 80% or above.

```
from pandas import DataFrame, read_csv


df: DataFrame = read_csv("math-students.csv")
```

- (a) Find the proportion (probability) of students who got an A in math.

```
# probability and proportion are not the same thing
A_s = df[df["G3"] >= 16]
print(f"percentage of math students who received an A: {round(len(A_s) / len(df) * 100, 1)}%")
```

```
percentage of math students who received an A: 10.1%
```

- (b) Find the proportion (probability) of students who had 10 or more absences.

```
# probability and proportion are not the same thing
oft_absent = df[df["absences"] >= 10]
print(f"percentage of math students who were absent 10 or more times: {round(len(oft_absent) / len(df) * 100, 1)}%")
```

```
percentage of math students who were absent 10 or more times: 21.0%
```

- (c) Find the probability of the intersection of the two events in (a) and (b).

```
# proportion, not probability:
oft_absent_A_s = oft_absent[oft_absent["G3"] >= 16]
print(f"percentage of math students who were absent 10 or more times and got an A: {round(len(oft_absent_A_s) / len(df) * 100, 1
```

```
percentage of math students who were absent 10 or more times and got an A: 1.3%
```

- (d) Find the probability of the union of the two events in (a) and (b).

```
# proportion, not probability:
union = [_ for _, row in df.iterrows() if row["G3"] >= 16 or row["absences"] >= 10]
print(f"percentage of math students who were absent 10 or more times or got an A: {round(len(union) / len(df) * 100, 1)}%")
```

```
percentage of math students who were absent 10 or more times or got an A: 29.9%
```

- (e) Find the conditional probability of a student getting an A grade if the student has 10 or more absences.

```
# proportion, not probability:
print(f"percentage of (math students who were absent 10 or more times) who got an A: {round(len(oft_absent_A_s) / len(oft_absent)
```

```
percentage of (math students who were absent 10 or more times) who got an A: 6.0%
```

On the difference between probability and proportion:

Huber, WA [Ignorance is Not Probability](). Risk Analysis Volume 30, Issue 3, pages 371–376, March 2010.